35 - Polymerisation

Types of polymerisation

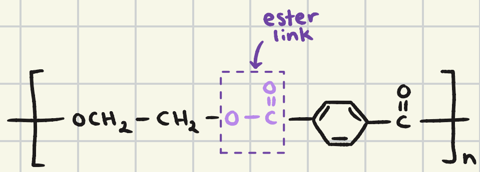
Feature	Addition Polymers	Condensation Polymers
Monomers used	Unsaturated monomers (alkenes with C=C double bonds)	Monomers with two functional groups (e.g., –COOH, –OH, $–NH_2$)
Type of reaction	Addition polymerisation	Condensation polymerisation
By-product formed	No by-product	Small molecules (e.g., water or HCl) are eliminated
Bond formation	Double bonds open to form long chains	Functional groups react to form ester or amide bonds
Examples of polymers	Poly(ethene), Poly(propene), PVC	Nylon, Terylene, Proteins, Starch
Biodegradability	Usually non-biodegradable	Some are biodegradable
Natural or synthetic	Mostly synthetic	Can be natural (e.g. proteins, starch) or synthetic (e.g. nylon)
Typical monomer examples	Ethene, Propene, Styrene	Dicarboxylic acid + diol or diamine, amino acids

Polyester

↳ a polymer with ester linkage

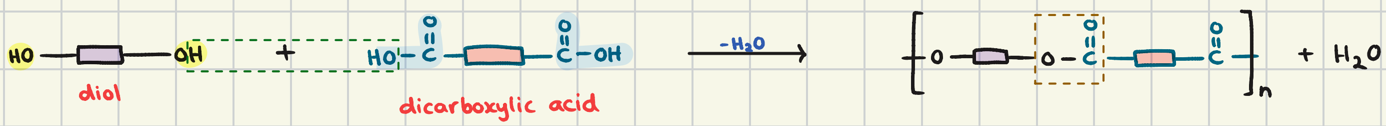
↳ formed via condensation reactions

e.g

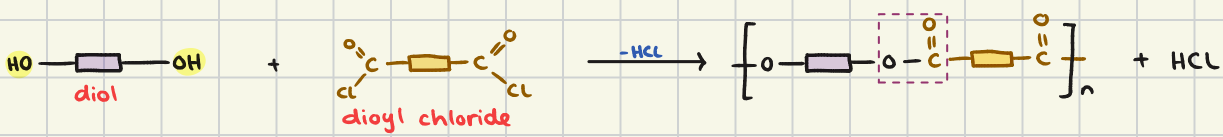


Formation of polyesters

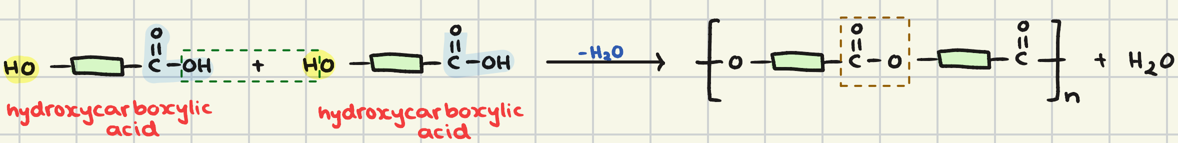
1) diol + dicarboxylic acid



2) diol + dioyl chloride

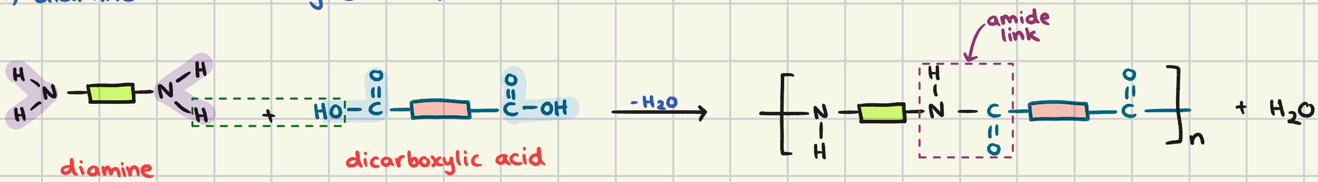


3) reaction of hydroxycarboxylic acid

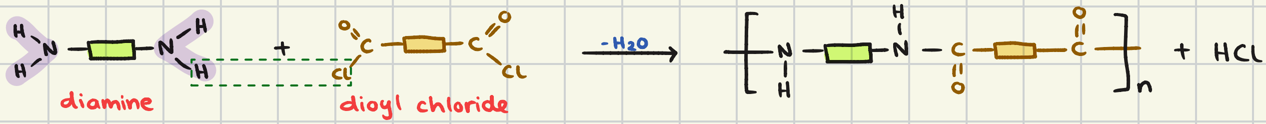


Formation of polyamides

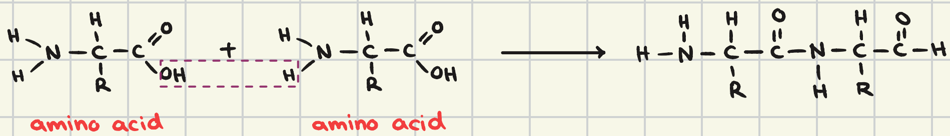
1) diamine + dicarboxylic acid



2) diamine + dioyl chloride



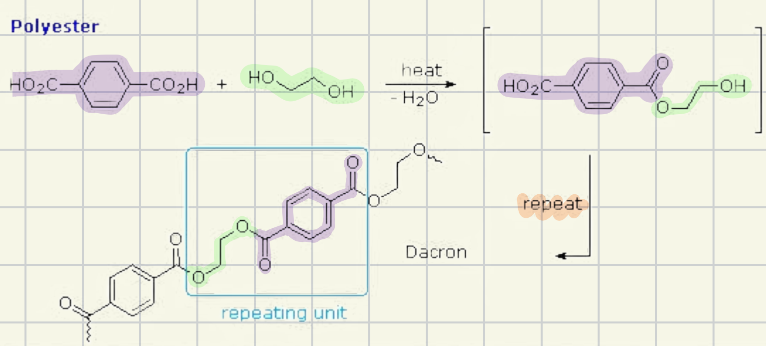
3) reaction of an $\overbrace{\text{aminocarboxylic acid}}^{\text{amino acid}}$ (similar to reaction of hydroxycarboxylic acid)

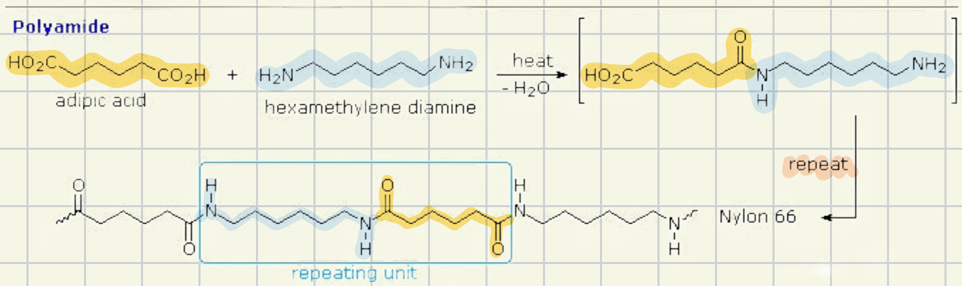


4) reaction b/w amino acids

SAME THING TBH (i think)

Deducing the repeat units of a condensation polymer

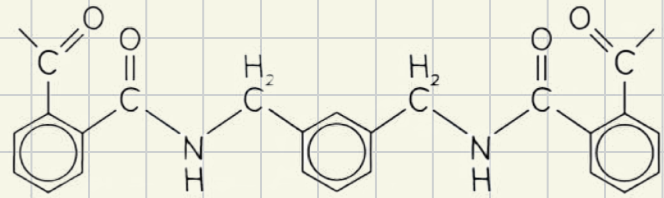




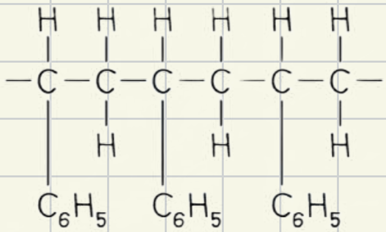
Worked example

Draw the repeating unit and identify the monomers used to make the following polymers

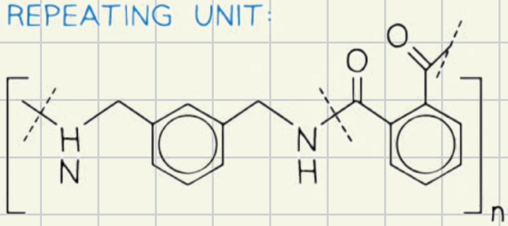
a)



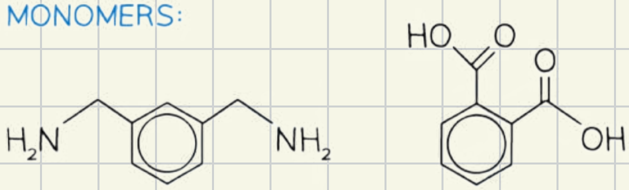
b)



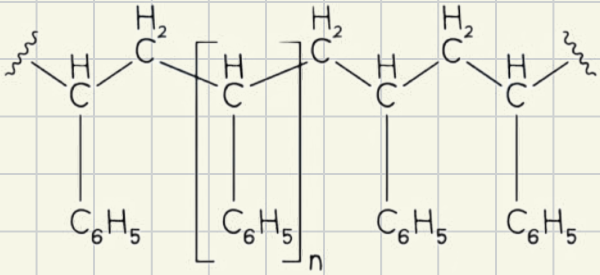
a) REPEATING UNIT:



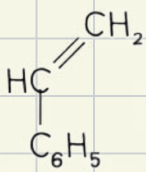
a) MONOMERS:



b) REPEATING UNIT:



b) MONOMERS:



Predicting the type of polymerisation

addition polymerisation

* requires monomers with double bonds
  e.g alkenes
* no by-products
* anything but ester/amide bond

condensation polymerisation

* requires monomers with two reactive functional groups
  e.g $-OH$, $-COOH$, $-COCl$, $-NH_2$
* small molecules like H_2O, HCl as by-products
* ester or amide linkage present

Degradeable polymers

→ poly (alkenes)

- chemically inert due to strong C-C & C-H bonds
- resistant to biodegradation, making them environmentally persistent

→ polymers degradeable by light

- some polymers can degrade under UV light exposure, breaking down into smaller fragments
  e.g PVC, polyethylene

→ biodegradeable polyesters & polyamides

- polyesters & polyamides can be broken down by hydrolysis (acidic or alkaline)

	Type of polymer	acidic hydrolysis	alkaline hydrolysis
Products	polyester	carboxylic acid + alcohol	carboxylate salt + alcohol
	polyamine	carboxylic acid + ammonium salt	carboxylate salt + amine